Bias Arcade

Teacher Guide

1. Overview

This guide is designed to help educators run a lesson on bias in AI systems using the Bias Arcade. It provides an overview of all available materials, a summary of the lesson structures and additional materials. No technical background is required to use this guide or to run the lessons. However, familiarity with how AI models work, such as how models are trained and learn from data, is helpful for getting the most out of the lessons. The materials and experiences are designed to be flexible. Educators can follow the suggested lesson structure closely or adapt it to fit their specific classroom context.

2. Material Overview

Documents	Description
Bias Types	Overview of all the bias types presented in FairAIED with case studies to illustrate how these biases manifest in real-world educational settings.
Teaching Resources	Collection of teaching materials including lesson plans and worksheets.
Single Lessons	Lesson Plans that focus on teaching one specific bias type in a single 1-hour lesson.
Jigsaw Lesson	Lesson Plan that focuses on teaching all four bias type in a single session.

3. Bias Type Overview

Bias type	Description
Learning Bias	Learning Bias frequently arises from modeling choices during training. Particularly the selection of objective functions that prioritize specific performance metrics might disadvantage individuals and groups.
Interaction Bias	Interaction Bias emerges from unequal engagement with a platform. Highly active users shape the algorithm's recommendations more strongly.
Confirmation Bias	Conformation Bias refers to the tendency to favor content, recommendations or outcomes that aligns with people's preexisting strengths, interests or behaviors limiting exposure to new subjects and challenges.
Algorithmic Bias	Algorithmic Bias typically originates from individual-level inconsistencies and propagates outward to reinforce broader group-level inequalities.

4. Lesson Overview

Bias Arcade offers two types of lesson formats to suit different classroom contexts and time constraints:

Single Lesson: Lessons built around one experience and one bias type. This format is ideal for educators who want to explore a specific bias in depth with time for gameplay, individual reflection and guided discussion. A separate lesson plan is available for each interactive experience.

Jigsaw Lesson: A longer session in which students are divided into four groups with each group playing a different interactive experience. In groups learners become “experts” on their bias types preparing themselves for teaching their bias type to the other learners. Expert groups are then split up and reorganized into new mixed groups each containing one member from every group. Students take turns sharing what they learned, ensuring every member of the new group is exposed to all bias types. This format encourages peer teaching, collaborative discussion, and a broader understanding of how bias manifests across different AI systems. A dedicated lesson plan with worksheets is available in Teaching Resources.

Learning Objectives

Lesson Type	Learning Objectives
Learning Bias	• Explain what learning bias is and how it can emerge during the AI training • Explain the role of data in training an AI model and how human choices in data selection change what the model learns. • Connect learning bias to a potential harmful outcome in a real-world context
Interaction Bias	• Explain what interaction bias is and how it can emerge • Identify underrepresented groups in a given situation and explain why they are absent
Confirmation Bias	• Define confirmation bias and explain how it operates in social media. • Identify inputs that feed recommendation algorithms (likes, reading time, scroll depth). • Describe the feedback loop between user behavior and feed recommendation algorithm.
Algorithmic Bias	• Define algorithmic bias and how it can emerge in AI systems. • Explain how an objective metric/goal might lead to unequal outcomes.
Jigsaw Lesson	• Explain their assigned bias to a peer using a concrete example. • Compare all four bias types.

	• Identify the source of each bias.

Prerequisites

No coding experience or technical knowledge is required. The games focus on concepts, consequences, and critical thinking rather than technical implementation. A basic understanding of how AI models work such as how they are trained and learn from data is helpful for getting the most out of bias arcade. Introductory materials for students who are unfamiliar with these concepts are available under Additional Resources.

Required Materials

- Computers or tablets with internet access
- Access to the Bias Arcade website (link)
- (Optional) Projector — useful for introducing the game to the whole class or displaying discussion questions during the debrief

5. Additional Resources

Building Prerequisites

The following resources can be used to introduce students to the basics of how AI models work before the lesson. These are particularly useful if students have little to no prior exposure to AI concepts.

- Game: AI Quests
  - Game-based learning experience designed to teach middle school students about AI
- Course: Elements of AI
  - Free course made for complete beginners by the University of Helsinki
- Game: AI for Oceans
  - Short interactive game where students train an AI to recognize fish vs trash in the ocean.

Diving Deeper

For students or educators who want to explore AI bias beyond the games, the following resources provide more in-depth coverage of the topic:

- Interactive Article: Searching for Unintended Biases With Saliency
  - Article that explores learning bias with watermarks
- Interactive Article: Confirmation Bias Test: Do You Only See What You Want to Believe?
  - Test that checks if you seek out information that challenges or supports your views
- Video: Beware online "filter bubbles" | Eli Pariser
  - Ted Talk on confirmations and filter bubbles
- Video: Algorithmic Bias in AI: What It Is and How to Fix It
  - Causes, Examples and Mitigation of Algorithmic Bias
- Reading: Surviving of the Best Fit Reading List
  - List for anyone who wants to learn more about the case for why and how we can make AI systems more inclusive